

Low-temperature Stress Alters Gene Expression of the Kidney Transcriptome
in *Takifugu rubripes*

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ABSTRACT

Cold temperature stress significantly impacts kidney gene expression, an important tissue for regulating the stress response in fish. *Takifugu rubripes* is an emerging aquaculture species important for the Chinese economy, but overwintering has severe adverse effects on growth performance. Understanding the mechanisms behind cold temperature stress will elucidate information about cold adaptation to aid in the breeding of cold tolerant strains. To investigate the effects of cold temperature stress on gene expression, kidney transcriptomes at 5 °C and 23 °C were analyzed, mapped, and compared using RNA-seq data and the STAR aligner. Using RStudio, the DESeq2 package identified 1388 differentially expressed genes (DEGs), with 583 upregulated and 805 downregulated. topGO enrichment analysis highlighted key pathways enriched during the stress response related to amino acid and lipid metabolism. Further analysis revealed downregulation of amino acid metabolism and both up and downregulation of lipid metabolism, suggesting a possible energy conservation strategy. Additional upregulation of genes related to membrane integrity, cell-cell adhesion, circadian rhythm, and apoptosis were observed, which likely assist in cold adaptation by promoting cellular communication, maintaining structural stability, creating an energy-efficient physiological state, and eliminating cell damage. This study was an important first step towards understanding cold adaptation of *Takifugu rubripes* at the transcriptional level. Further research would overcome the present limitations, possibly through a multi-omics approach or machine learning models, to provide more mechanistic insights into understanding organismal survival adaptations.

INTRODUCTION

Environmental stress can impact an organism at the molecular level, resulting in differentially expressed genes. The change in expression impacts biological features like metabolism, cytoskeletal organization, and proteolysis, allowing the organism to adapt to the stressor (De Nada, 2011). In particular, at low temperatures, metabolism is severely altered to increase the energy available for an organism to allow the continuous function of biological mechanisms imperative to survival (Xu, 2018). Protein synthesis consumes one third of all energy produced by the cell. To account for the energy demand of survival mechanisms, translation is downregulated, and growth is hindered (Hernández-Elvira, 2022).

During a stress response, the central nervous system (CNS) will recognize the potential stressor, leading to a cascade of events that results in the release of the corticotropin-releasing hormone from the hypothalamus, which stimulates the release of the adrenocorticotrophic hormone from

the pituitary, which finally results in the release of cortisol from adrenal glands (Barton, 2002). Glucocorticoid receptors act as ligand-inducible transcription factors and selectively activate or repress genes in response to cortisol binding (Bury, 2007). Kidneys are part of the hypothalamic-pituitary-interrenal axis and contain adrenal glands to secrete cortisol into the bloodstream (Liu, Z., 2022). In addition, kidneys are also important for amino acid metabolism - involving synthesis, reabsorption, and excretion - and lipid metabolism - involving fatty acid utilization and cholesterol control (Khan, 2013). As such, studying the change in gene expression of kidneys is crucial for understanding an organism's response to stress.

China has become one of the largest producers of aquaculture, a flourishing industry that is exceedingly diverse in species. With rapid growth yields and technological advances, the integration of this industry into global food systems has expanded and will have significant implications in the future to sustain global food demands (Hou, 2022). However, environmental conditions during the winter can severely impact the growth and survival of fish when sea water temperatures decrease. To rectify this, fish are often brought indoors to specialized heating pools, but temperatures are still kept relatively low for economic and environmental reasons. The generation of cold tolerant strains is significant to increase survival rates and reduce the impacts on the economy and aquaculture industry (Liu, Z., 2022).

Takifugu rubripes, or the tiger puffer, is a species rich in collagen and protein, but its consumption was banned due to the tetrodotoxin contained within its body. However, the development of aquaculture technology established the tiger puffer safe for consumption, rendering China as one of its major exporters (Hou, 2022). This species is eurythermal, with a growth-temperature range between 16 - 25 °C, and suitable for cage culture, making it an ideal model organism for studying the impacts of temperature stress. At a temperature of 5 °C, the

tiger puffer enters a state of cold shock but can recover when a normal temperature is restored (Liu, Z., 2022). Thus, studying the mechanisms behind cold stress is feasible at this extreme temperature to identify key pathways and genes responsible for cold adaptation in *Takifugu rubripes*.

The aim of this study was to investigate the effects of cold temperature stress on the expression of kidney genes in *Takifugu rubripes*. Kidney transcriptome analysis was performed under normal (23 °C) and cold (5 °C) temperature conditions using RNA-seq data generated from Liu, Z., et al. Based on data from the literature, treatment of *Takifugu rubripes* with low temperatures in comparison to the control should result in the downregulation of kidney genes associated with amino acid and lipid metabolism.

MATERIALS AND METHODS

Experimental fish and sample collection

Information on the tiger pufferfish sample selection was obtained from the paper, “Kidney transcriptome analysis of tiger puffer (*Takifugu rubripes*) uncovers key pathways and genes in response to low-temperature stress” written by Liu, Z., et al. in 2022.

A total of 360 healthy and vigorous tiger pufferfish juveniles were obtained from full-sibling families established at a local fish farm in Yantai, China. The choice of siblings was made to reduce the effect of genetic variation on the outcome of the RNA-seq analysis. The selected juveniles had an average length of 6.2 ± 0.2 cm and a body weight of 5.1 ± 0.5 g. The fish were separated into six 600L tanks, three tanks for the control group (23 °C) and three tanks for the low temperature treatment group (5 °C).

For 21 days prior to the experiments, the fish were acclimated to the culture conditions (23 ± 0.5 °C, 30 ± 0.2 ppt, natural photoperiod about 14 h day and 10 h night). Water conditions were as follows: pH, 7.80–7.86; dissolved oxygen, > 6 mg/L; nitrites, 0.05–1.20 mg/L; nitrates, 3.20–20.10 mg/L; and ammonium, 0–0.2 mg/L. Once acclimated, they were subjected to two different treatments; the control treatment (23 °C) and the low temperature treatment (5 °C). The low temperature treatment was administered to three tanks by lowering the tank temperature (23 °C) by 3 °C each day until the desired temperature (5 °C) was reached. Six individuals from each tank were randomly selected and anesthetized with 200 mg/L MS222. The kidney tissue of the individuals was immediately isolated on ice and frozen with liquid nitrogen to await RNA extraction and analysis.

RNA extraction and kidney transcriptome sequencing

As described by Liu, Z., et al., a total RNA was extracted from the kidney tissue using TRIzol reagent, according to the Invitrogen protocol (Invitrogen, USA), and treated with RNase-free DNase I to remove genomic DNA. RNA quality and quantity were determined using a 2100 Bioanalyser and ND-2000, respectively. The six samples from each tank had their extracted RNA combined into a single sample for further analysis. 1 µg of total RNA from each combined sample was used to prepare a mRNA-Seq library, using a TruSeq RNA Sample Prep Kit (Illumina, USA) according to the manufacturer's instructions. Libraries were size-selected for cDNA target fragments of 200–300 bp on 2 % Low-Range Ultra Agarose, followed by amplification by polymerase chain reaction (PCR) using Phusion DNA polymerase (New England Biosciences, USA) for 15 PCR cycles. Amplified library preparations were quantified using a TBS380 fluorometer. Preparations were then sequenced with the Illumina NovaSeq platform (2×150 bp read length).

Data and Computational Tools

The reference genome for *Takifugu rubripes* (NCBI RefSeq Assembly: GCF_901000725.2), annotation file (.gtf) and gene association file (.gaf) were obtained from the “Genome assembly fTakRub1.2” page on the NCBI website. The RNA-seq data (SRA: PRJNA739387) files for the control and low-temperature treatment, generated by Liu, Z.,, were obtained from the Sequence Read Archive (SRA) on the NCBI website.

The RNA-seq analysis was performed using Bash and R (version 4.4.2) in the RStudio environment, with FastQC for quality control and STAR (Spliced Transcripts Alignment to a Reference) for sequence alignment.

Quality control and sequence alignment

The *Takifugu rubripes* reference genome, annotation file (.gtf) and RNA-seq data files were uploaded to the terminal for analysis. FastQC was performed on the control RNA-seq files to assess the quality of the data. A genome index was generated from the reference genome and annotation files using STAR. The read length used was 151 bp and the number of threads to be used for genome generation was eight. STAR was then used to align the RNA-seq data to the newly generated genome index. The number of threads used for alignment was 16 and the quantification mode used was GeneCounts. The file containing the read counts per gene (ReadsPerGene.out.tab) file that resulted from each alignment was downloaded from the terminal and used for further analysis.

Count normalization and differential gene expression analysis

RStudio was used to run R, with the following packages loaded: tidyverse for data manipulation, DESeq2 for differential gene expression analysis, and pheatmap, ggplot2, patchwork, and ggrepel for plotting the results.

The read counts per gene file (.tsv) were loaded into the environment and compiled into a matrix. A metadata file containing the column information for the matrix was also created. The DESeq data set was created from the matrix and metadata file. The independent variable designated for investigation was pair + condition to ensure that each control dataset was compared to its corresponding low temperature dataset (ex. control1 vs. low_temp1). DESeq2 was run to calculate differential expression and the resulting DESeq DataSet (dds) object was used for further analysis. PCA on log-transformed count data and sample-to-sample distances were generated for identifying similarities and differences between the samples.

Total reads per sample, group-specific reads, and overall totals were calculated. Exploration of missing values (NAs) in the results was performed. An MA plot is created to visualize the relationship between the mean of normalized counts in log scale and the log2 fold change of genes.

Differentially Expressed Genes

The results of the comparison between the low temperature and control group were extracted from the dds object as a dataframe. For the analysis of differentially expressed genes, we focused on those with an adjusted p-value (padj) below or equal to 0.05, indicating statistical significance. Only the genes with two times higher (upregulated genes with $\log_2\text{FoldChange} \geq 1$) or lower (downregulated genes with $\log_2\text{FoldChange} \leq -1$) expression were deemed biologically

relevant and kept. Genes with NA values were filtered. From the resulting dataset, the top 10 upregulated and downregulated genes were identified and stored in different .csv files. The dataset with all the differentially expressed genes was saved in .csv format, which is used in gene ontology (GO) enrichment analysis.

Heatmaps of z-scores are generated for these top 10 upregulated and downregulated genes, showing their standardized expression patterns across samples after variance stabilizing transformation (VST). Finally, visualization of the differentially expressed genes distribution and plots of normalized gene expression counts for the top 10 upregulated and downregulated genes are performed.

Gene ontology enrichment term analysis

The R packages tidyverse for data manipulation and topGO for performing GO term enrichment analysis were loaded. The GO terms mapping file (.tsv) for associating the genes with their GO terms was generated through the manipulation of the gene association file (.gaf). Because the Symbol column of the .gaf file corresponds to the gene IDs of our read counts file, the .gaf file was collapsed such that for each gene, all its GO terms were stored in the same row (rather than in different rows such as they are display in the original .gaf file).

GO enrichment analysis was performed for both the upregulated and downregulated genes using topGO and the generated file that links gene IDs to GO terms. topGOdata objects are created specifying the Biological Process (BP) ontology and linking genes to their GO terms. Using Fisher's exact test with the "weight01" algorithm, it calculates enrichment scores to identify GO terms significantly associated with the upregulated and downregulated gene sets.

Visualization of both downregulated and upregulated genes results was performed. For both datasets, it filters the top 20 GO terms based on significance ($p\text{-value} \leq 0.05$), calculates the Gene Ratio, and arranges the terms by their enrichment ratio. The combination of the upregulated and downregulated GO enrichment results was stored in a dataframe for filtering significant terms ($p\text{-value} \leq 0.05$) and calculating the Gene Ratio. The top 40 terms, arranged by Gene Ratio, are visualized in a faceted scatterplot.

The GitHub repository containing the files and codes associated with this project is available in Supplementary Note 1.

RESULTS

To investigate the gene expression profiles in tiger puffer kidney in response to low-temperature challenges, a total of 137,010,065 reads were collected, with 67,958,297 reads (49.6%) in the control group and 69,051,768 reads (50.4%) in the low-temperature group (Supplementary Tables 1, 2).

FastQC analysis revealed warnings related to GC content in the samples, which is consistent with the high GC content of the *Takifugu rubripes* transcriptome (Chaurasia, 2014). Read lengths were consistent across all sequences in all samples, and while some adapter content was detected, overrepresented sequences were listed as 'No Hit,' indicating no matches to common contaminants. The expected failure in sequence duplication levels reflects typical transcriptome data with highly abundant and lowly abundant transcripts. All reads were deemed high quality and included in the analysis (Supplementary Figure 9).

Of the total reads, 56,961,067 (41.6%) were successfully mapped to the reference genome, maintaining similar proportions between the control (49.4%) and low-temperature groups (50.6%), consistent with the distribution of total raw reads (Supplementary Tables 1, 2).

PCA of gene expression data obtained with DESeq2 shows clear separation between low-temperature and control groups along PC1, which explains 86% of the variance, suggesting that variation in the dataset is driven by the experimental condition. PC2 accounts for 8% of the variance, representing biological variability between replicates or noise in the data. Tight clustering of the low-temperature samples suggests consistency, while one control sample, *Control2*, appears to be a potential outlier (Figure 1a).

The sample-to-sample distance heatmap confirms the observation from the PCA plot, suggesting that *Control2* is likely an outlier due to either biological variability, technical artifacts, or preprocessing issues (Figure 1b). However, after a careful review of the RNA extraction, library preparation, and transcriptome sequencing protocols of the samples, no evidence suggests a specific failure or deviation affecting *Control2* (Liu Z., 2022). Additionally, FastQC analysis showed high sequencing quality and consistency across all samples, and library sizes are relatively consistent, ranging from ~9.0M to ~9.8M reads (Supplementary Figure 9, Supplementary Table 2).

While all fish came from the same full-sibling family, minor genetic differences could affect gene expression, and individual fish may exhibit variable physiological responses even under carefully controlled conditions (Liu L., 2020). Thus, the behavior of *Control2* is most likely due to biological variability rather than technical artifacts or preprocessing issues. All samples were

included in the analysis given that DESeq2 is designed to handle biological variability and accounts for differences between replicates statistically (Love, 2014).

The MA plot provides a visual representation of the analyzed genes, with statistically significant genes highlighted in blue. Upregulated genes are shown as blue points above the green dotted line, which corresponds to a $\log_2\text{FoldChange}$ of 1. Downregulated genes are shown as blue points below the red dotted line, which corresponds to a $\log_2\text{FoldChange}$ of -1 (Figure 2a).

Some genes show missing values for their adjusted p-values, labeled as 'NA' (not available) in the MA plot (Figure 2a). In this dataset, these genes correspond to cases where DESeq2 is unable to calculate p-values or $\log_2$ fold changes due to the baseMean column being equal to zero (Supplementary Table 3). The baseMean column in DESeq2 objects represents the average normalized expression of a gene across all samples. If a gene has no detectable expression in any sample, its baseMean will be zero, making it impossible to calculate fold changes or perform statistical testing (Love, 2014). Additionally, even for genes where DESeq2 calculates p-values, some may lack adjusted p-values (padj) because these are derived after correcting for multiple testing (Supplementary Table 3). Genes with extremely low counts or unreliable estimates are often filtered out during this step, resulting in unassigned adjusted p-values (Love, 2014).

A total of 1,388 differentially expressed genes were identified, with 583 (42%) being upregulated and 805 (58%) downregulated. (Figure 2b). Among the upregulated genes, LOC101074542 is associated with hormone and progesterone metabolism, while LOC101063073 regulates transcription and circadian rhythm, potentially impacting metabolic homeostasis. Similarly, in the downregulated genes, LOC115248888 is linked to long-chain fatty acid metabolism, suggesting its role in lipid processing and mitochondrial function, and LOC101078510 plays a

critical role in amino acid catabolism, specifically L-serine and threonine breakdown, enabling pyridoxal phosphate binding (Table 1). Genes related to amino acid metabolism (*LOC101078510*, *LOC115248888*) and lipid metabolism (*LOC115248888*) are downregulated (Figure 3a). A notable downregulation of genes is observed in *Control2* (Figure 3c), highlighting biological variation, which is also evident in the normalized counts (Supplementary Figure 1).

The GO term enrichment analysis was used to identify the functions of the DEGs (Figure 4). Biological processes associated with lipid and amino acid metabolism, which are essential for energy production and homeostasis, were downregulated, reflecting a potential conservation strategy. Simultaneously, upregulated pathways related to biosynthesis, signaling, and immune responses underscore the organism's active adaptation to mitigate cold stress. Individual plots reveal that a large number of downregulated genes are associated with highly significant biosynthetic and organic acid metabolic processes, along with a high enrichment ratio for the acetyl-CoA biosynthetic process from acetate (Supplementary Figure 3). Conversely, a large number of upregulated genes are associated with transmembrane transport, protein phosphorylation, and cell-cell adhesion, with a moderate enrichment ratio for the long-chain fatty-acyl-CoA metabolic process and a high enrichment ratio for the negative regulation of circadian rhythm (Supplementary Figure 4).

Differential gene expression analysis, excluding *Control2* and *Low2* samples due to their high sample-to-sample distance, showed improved clustering in PCA and reduced distances in the sample matrix (Supplementary Figures 6). However, GO term enrichment analysis revealed similar pathways, reaffirming that the variation of *Control2* is biological and its removal does not significantly affect DEG associated pathways (Supplementary Figures 6-8).

DISCUSSION

Understanding how transcriptomic processes shift in response to cold stress in a model organism provides invaluable insights into the biological and genomic mechanisms underpinning environmental adaptation. In this study, we conducted differential gene expression analysis and GO term enrichment on RNA-seq samples of the kidney transcriptome from Japanese tiger pufferfish (*Takifugu rubripes*) exposed to normal (23°C) and cold (5°C) temperatures. We hypothesized that lower temperature environments would downregulate kidney metabolism genes, particularly those associated with amino acid and lipid metabolism, as an energy-preservation strategy (Liu Z., 2022). Our findings support part of our hypothesis, revealing statistically significant ($p < 0.006$) downregulation of genes involved in amino acid and confounding evidence of both up and down regulation in long-chain fatty acid metabolism (Figure 4, Supplementary Figure 3, 4). Additionally, we observed upregulation of genes related to membrane integrity, cell-cell adhesion, and circadian rhythm; shedding light on molecular mechanisms that help these organisms adapt to cold temperatures.

GO term enrichment highlighted specific down regulated pathways, including processes related to quinolinate (a metabolite of L-tryptophan), L-phenylalanine, tyrosine, and glutathione metabolism (Supplementary Figure 3). These findings suggest that suppression of amino acid metabolism may be crucial for maintaining homeostasis during cold stress. Energy-intensive amino acid metabolism is likely downregulated as an adaptive response to food scarcity, which is common in colder conditions (Wen, 2019). Reduced protein synthesis demand further diminishes the need for amino acid catabolism, which would otherwise generate precursor molecules (Liu L., 2020).

A unique aspect of cold adaptation in kidney function is the reduced nitrogen waste generation. Amino acid breakdown produces nitrogenous byproducts that can accumulate under conditions of impaired renal function due to cold stress (Ma, 2021). Supporting this, we identified the *rhcg2* gene, a regulator of nitrogen levels, as one of the top 10 downregulated genes (Table 1). Thus, the downregulation of amino acid metabolism in the pufferfish appears to be driven by a combination of energy preservation, decreased demand for metabolites, and waste management strategies.

Our results reveal nuanced changes in lipid metabolism under cold stress. While Figure 4 demonstrates a general decrease in long-chain fatty acid metabolism, Supplementary Figure 4 highlights concurrent upregulation of specific lipid-related pathways, suggesting differential regulation across genes involved in lipid metabolism. Such variability may reflect the dual role of lipids during stress: while energy conservation and membrane fluidity maintenance necessitate downregulation, lipids can also serve as an efficient energy source under extreme conditions (Los, 2013). For example, the *LOC115248888* gene, involved in fatty acid metabolism, was identified as significantly downregulated, supporting the trend of energy conservation. Additionally, lipid metabolism may still be required as it is closely linked to maintaining membrane integrity (Ammendolia, 2021). Thus, stress-induced reliance on stored glycogen and lipids for energy may drive upregulation in alternate lipid-related pathways, reflecting the multifaceted nature of cold adaptation.

Cold environments significantly impact cell membrane integrity, making membranes more rigid and impairing vital cellular processes (Los, 2013). To counteract these effects, our analysis revealed upregulation of genes enhancing cell-cell signaling, adhesion, and regulation of cell shape. These adaptations likely optimize tissue functionality by promoting cellular

communication and maintaining structural stability. Notably, the prevention of ectodomain proteolysis ensures intact extracellular signaling, which is crucial for survival in cold stress conditions (Supplementary Figure 4).

An intriguing finding was the negative regulation of circadian rhythm, which may help prevent metabolic imbalances under cold stress (Bass, 2010). Circadian rhythm disruption could reduce metabolic fluctuations and facilitate a stable, energy-efficient physiological state, further contributing to cold adaptation mechanisms. Another notable detail was the upregulation of genes involved in apoptosis (Table 1). It is possible that cold stress leads to dysregulation of cellular processes, and programmed cell death is a method to eliminate damaged cells. Stress responses, like those caused by colder temperatures can activate tumor suppressor gene p53, leading to downstream cell cycle arrest and apoptosis (Vousden, 2007).

This study has several limitations rooted in the current scope of gene annotations and the inherent constraints of GO term enrichment analysis. Many genes, including prominently downregulated ones such as *LOC101063513*, remain uncharacterized, limiting our understanding of their functional roles and their broader contributions to transcriptomic shifts under cold stress. This lack of annotation creates gaps in deciphering how various genes interact to benefit cellular responses and survival. Furthermore, the broad categorizations provided by GO analysis often fail to capture the nuances of specific cellular processes, making it challenging to establish precise links between transcriptional changes and physiological adaptations.

Biases in GO term enrichment analysis also warrant consideration. Well-characterized gene types are often prioritized, potentially overshadowing the significance of less-annotated genes. Additionally, reliance on enrichment ratios alone can be misleading, particularly when the

number of significant genes is small ($25 >$). Smaller gene sets can artificially inflate enrichment ratios due to a low denominator, underscoring the importance of considering both the ratio and the total number of significant genes to ensure robust conclusions.

Future research should aim to overcome these limitations through integrative and innovative approaches. A multi-omics strategy, combining genomic, proteomic, and metabolomic data, could provide a more comprehensive understanding of the pathways involved in cold-stress responses in *Takifugu rubripes*. This approach would allow researchers to validate transcriptomic findings at multiple levels and reveal deeper insights into specific mechanisms of adaptation.

Moreover, machine learning models, such as decision trees and random forests, offer promising avenues for data analysis. These algorithms can process large and complex transcriptomic datasets, uncovering patterns and relationships that might otherwise go unnoticed (Pouyan, 2018). Such tools could help identify critical genes and regulatory networks that drive cold-stress responses, paving the way for more targeted investigations.

Overall, results show that downregulated genes predominantly suppress pathways related to amino acid and lipid metabolism, consistent with a resource-conservation strategy, while upregulated genes enhance adaptive and stress-responsive pathways. This dual perspective highlights the dynamic nature of transcriptomic responses to environmental challenges

Our findings also suggest an intriguing link between metabolic regulation and circadian rhythms, which merits further exploration. Investigating how environmental stressors like cold influence circadian gene expression could uncover novel strategies organisms use to maintain homeostasis.

As gene annotation databases continue to expand and improve, GO enrichment analyses will become increasingly precise and reliable, enhancing our ability to draw connections between transcriptional changes and cellular adaptations. This study serves as a step toward understanding the transcriptomic shifts underpinning cold adaptation in *T. rubripes*. With continued advancements in technology and annotation resources, future research can build on these findings to uncover the intricate strategies marine organisms employ to survive in extreme environments, offering broader implications for understanding resilience across species.

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REFERENCES

- Ammendolia, D. A., Bement, W. M., & Brumell, J. H. (2021). Plasma membrane integrity: implications for health and disease. *BMC Biology*, 19(1). <https://doi.org/10.1186/s12915-021-00972-y>
- Barton, B. A. (2002). Stress in Fishes: A Diversity of Responses with Particular Reference to Changes in Circulating Corticosteroids. *Integrative and Comparative Biology*, 42(3), 517–525. <https://doi.org/10.1093/icb/42.3.517>
- Bass, J., & Takahashi, J. S. (2010). Circadian integration of metabolism and energetics. *Science*, 330(6009), 1349–1354. <https://doi.org/10.1126/science.1195027>
- Bury, N. R., & Sturm, A. (2007). Evolution of the corticosteroid receptor signalling pathway in fish. *General and Comparative Endocrinology*, 153(1–3), 47–56. <https://doi.org/10.1016/j.ygcen.2007.03.009>
- Chaurasia, A., Tarallo, A., Berna, L., Yagi, M., Agnisola, C., et al. (2014) Length and GC Content Variability of Introns among Teleostean Genomes in the Light of the Metabolic Rate Hypothesis. *PLoS ONE* 9(8): e103889. <https://doi.org/10.1371/journal.pone.0103889>
- De Nadal, E., Ammerer, G., & Posas, F. (2011). Controlling gene expression in response to stress. *Nature Reviews Genetics*, 12(12), 833–845. <https://doi.org/10.1038/nrg3055>
- Hernández-Elvira, M., & Sunnerhagen, P. (2022). Post-transcriptional regulation during stress. *FEMS Yeast Research*, 22(1). <https://doi.org/10.1093/femsyr/foac025>
- Hou, H., Zhang, Y., Ma, Z., Wang, X., Su, P., Wang, H., & Liu, Y. (2022). Life cycle assessment of tiger puffer (*Takifugu rubripes*) farming: A case study in Dalian, China. *The Science of the Total Environment*, 823, 153522. <https://doi.org/10.1016/j.scitotenv.2022.153522>
- Khan, K. N. M., Hard, G. C., & Alden, C. L. (2013). Kidney. In *Elsevier eBooks* (pp. 1679–1685). <https://doi.org/10.1016/b978-0-12-415759-0.00047-9>
- Liu, L., Zhang, R., Wang, X., Zhu, H., & Tian, Z. (2020). Transcriptome analysis reveals molecular mechanisms responsive to acute cold stress in the tropical stenothermal fish tiger barb (*Puntius tetrazona*). *BMC Genomics*, 21(1). <https://doi.org/10.1186/s12864-020-07139-z>
- Liu, Z., Ma, A., Yuan, C., Zhu, L., Chang, H., Yu, L., & Meng, X. (2022). Kidney transcriptome analysis of tiger puffer (*Takifugu rubripes*) uncovers key pathways and genes in response to low-temperature stress. *Aquaculture Reports*, 26, 101287. <https://doi.org/10.1016/j.aqrep.2022.101287>
- Los, D. A., Mironov, K. S., & Allakhverdiev, S. I. (2013). Regulatory role of membrane fluidity in gene expression and physiological functions. *Photosynthesis Research*, 116(2–3), 489–509. <https://doi.org/10.1007/s11120-013-9823-4>
- Love, M.I., Huber, W. & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol* 15, 550. <https://doi.org/10.1186/s13059-014-0550-8>
- Ma, P., Yin, X., Li, D., Li, L., & Tang, R. (2021). Acute temperature changes induce an oxidative stress response in kidney cells of grass carp *Ctenopharyngodon idellus*. *Fisheries Science*, 87(6), 775–784. <https://doi.org/10.1007/s12562-021-01545-2>
- Pouyan, M. B., & Kostka, D. (2018). Random forest based similarity learning for single cell RNA sequencing data. *Bioinformatics*, 34(13), i79–i88. <https://doi.org/10.1093/bioinformatics/bty260>

- Vousden, K., Lane, D. p53 in health and disease. *Nat Rev Mol Cell Biol* 8, 275–283 (2007).
<https://doi.org/10.1038/nrm2147>
- Wen, X., Hu, Y., Zhang, X., Wei, X., Wang, T., & Yin, S. (2019). Integrated application of multi-omics provides insights into cold stress responses in pufferfish *Takifugu fasciatus*. *BMC Genomics*, 20(1). <https://doi.org/10.1186/s12864-019-5915-7>
- Xu, D., You, Q., Chi, C., Luo, S., Song, H., Lou, B., & Takeuchi, Y. (2018). Transcriptional response to low temperature in the yellow drum (*Nibea albiflora*) and identification of genes related to cold stress. *Comparative Biochemistry and Physiology Part D Genomics and Proteomics*, 28, 80–89. <https://doi.org/10.1016/j.cbd.2018.07.003>

FIGURES AND TABLES

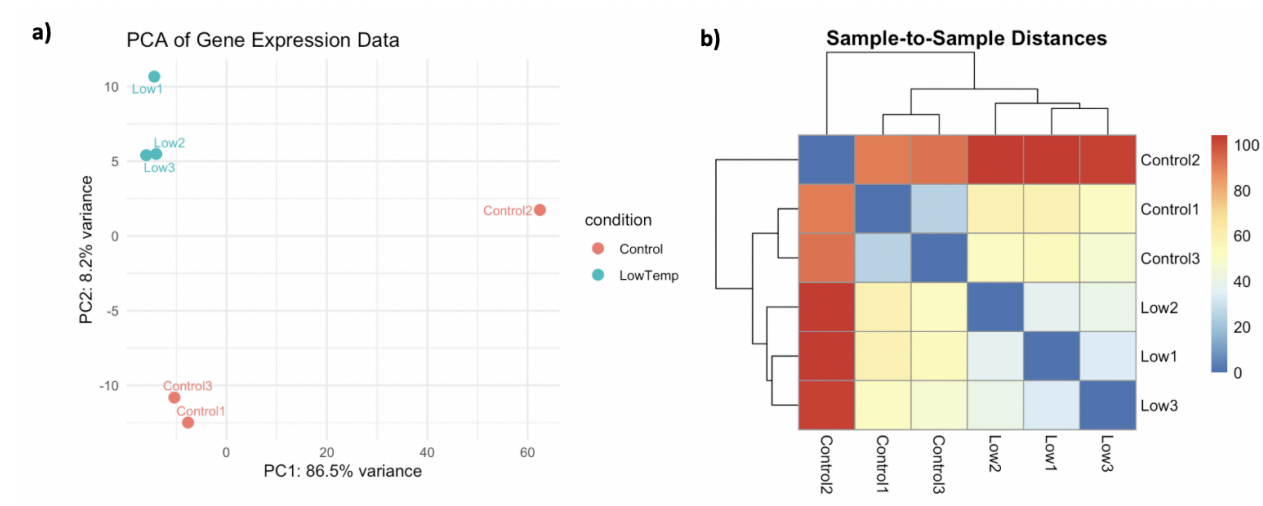


Figure 1: **a)** Principal Component Analysis (PCA) plot of gene expression data. Samples are colored by group. **b)** Heatmap of sample-to-sample distances. Euclidean distances between normalized gene expression profiles (rlog-transformed) were taken. Clustering of the rows and columns reveals similarities between samples.

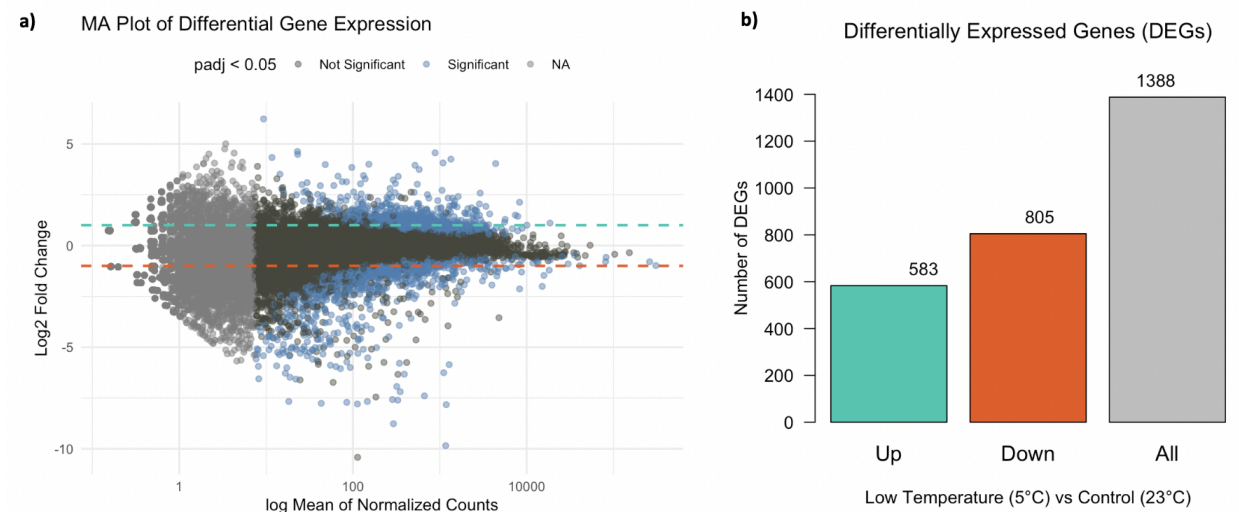


Figure 2: a) MA Plot of differential gene expression analysis showing the relationship between the mean normalized expression counts in log scale and the log2 fold changes in gene expression for all genes with a defined adjusted p-value (padj). Significant genes ($\text{padj} < 0.05$) are highlighted in blue. Dashed lines account for the log2 fold change thresholds for upregulated (1) and downregulated (-1) genes. **b).** Number of Differentially Expressed Genes (DEGs) between the low-temperature and control groups, categorized as upregulated (Up), downregulated (Down), and the total of DEGs (All).

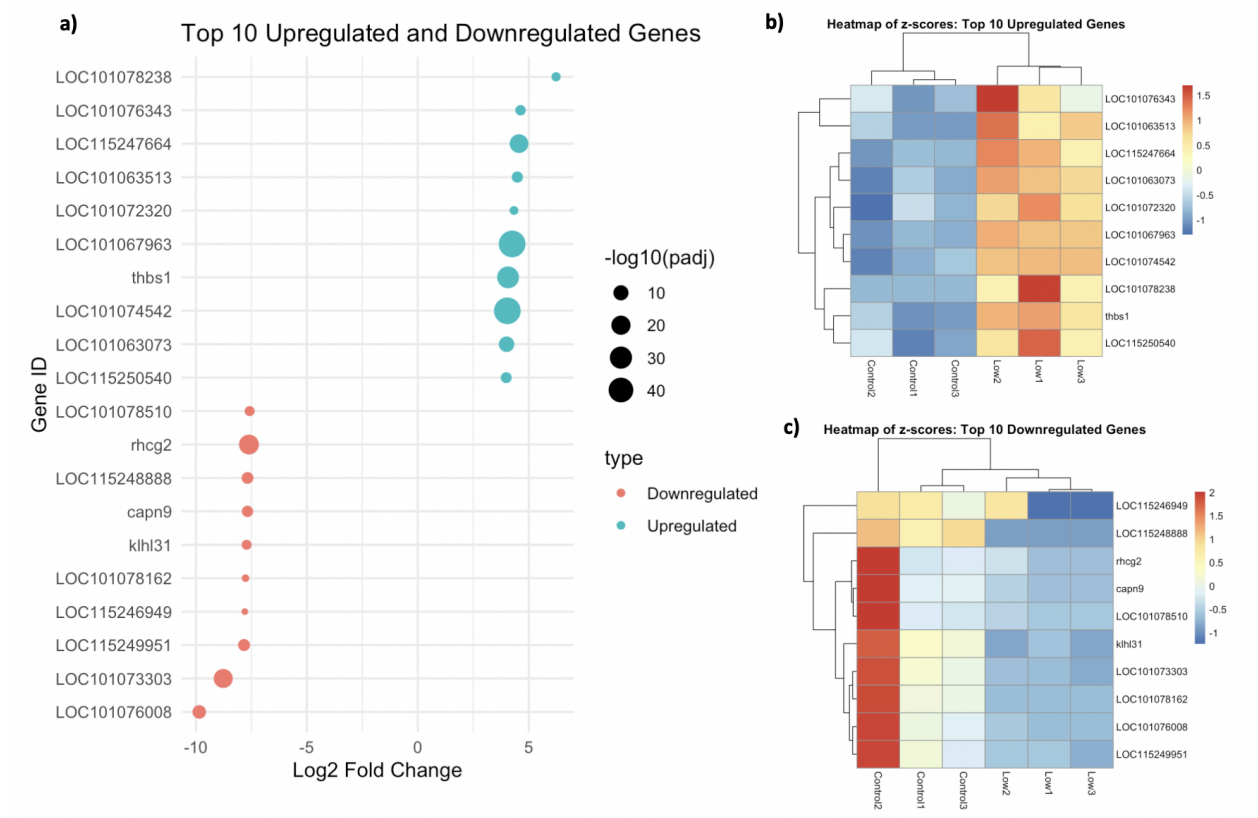


Figure 3: **a)** Top 10 upregulated and downregulated genes in *Takifugu rubripes* kidney under low-temperature stress. Dot size indicates statistical significance. **b)** and **c)** provide heatmaps of z-scores for the top 10 upregulated and downregulated genes respectively, showing clustering patterns of gene expression profiles across control and low-temperature conditions.

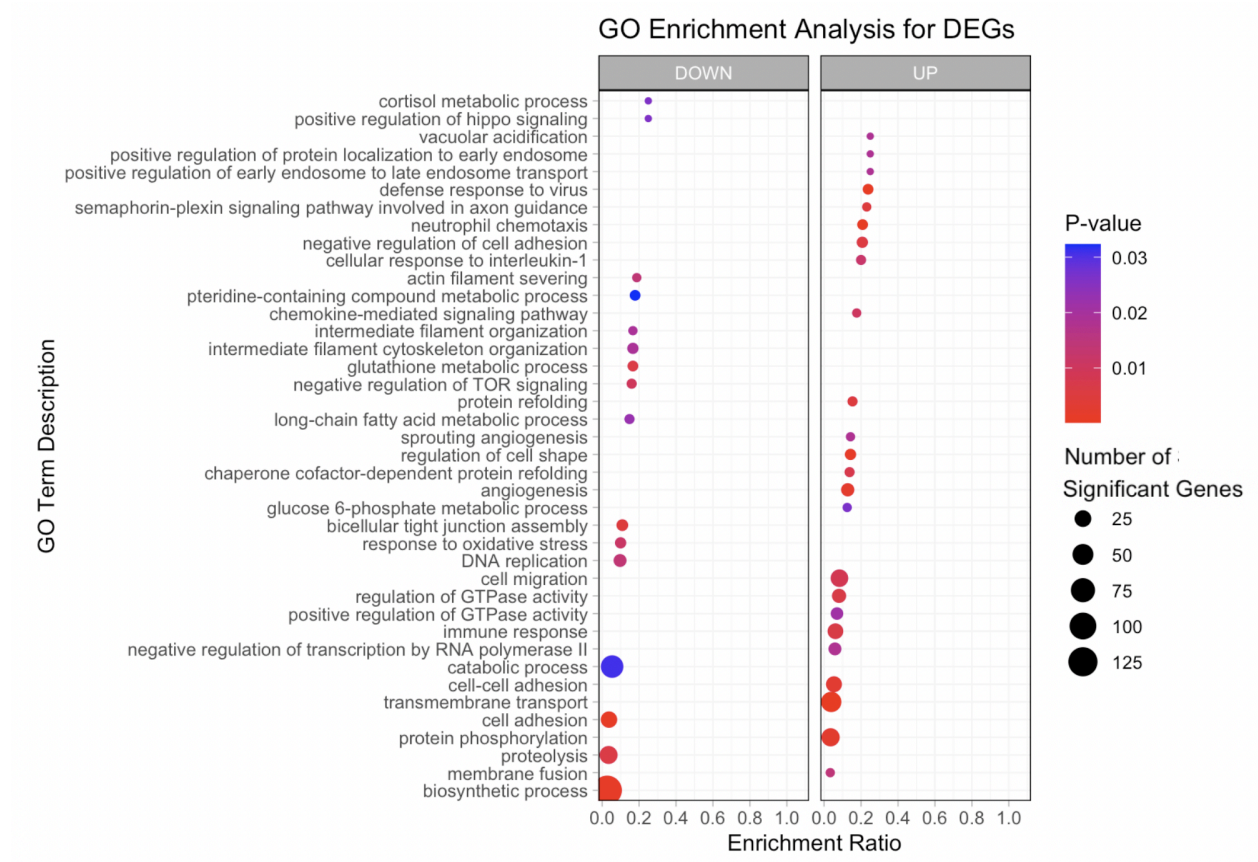


Figure 4. GO enrichment analysis for DEGs, displaying the top 40 GO term descriptions separated by downregulated (Down) and upregulated (Up) genes. The size of each dot corresponds to the number of significant genes associated with the term, while the color represents the p-value, with more statistically significant terms highlighted in red.

Table 1. Summary of the top 10 upregulated and downregulated genes from the NCBI Gene database.

Top 10 Upregulated Genes		Top 10 Downregulated Genes	
Gene ID	Summary	Gene ID	Summary
LOC101078238	Apoptosis execution, located in plasma membrane.	LOC101076008	Involved in chromatin structure, cell survival, and immune response; located in the extracellular matrix.
LOC101076343	Membrane protein; function unspecified.	LOC101073303	Uncharacterized protein; located in extracellular space.
LOC115247664	Phosphate transmembrane transporter.	LOC115249951	Structural molecule activity protein; function unspecified.
LOC101063513	Uncharacterized protein.	LOC115246949	Uncharacterized protein.
LOC101072320	Structural cytoskeleton protein, involved in filament organization and autophagy; located in cytoplasm and cell projection.	LOC101078162	Protein binding in extracellular matrix and space.
LOC101067963	Membrane protein; function unspecified.	klhl31	Involved in negative regulation of JNK cascade and negative regulation of protein phosphorylation.
thbs1	Adhesive glycoprotein that mediates cell-to-cell and cell-to-matrix interactions.	capn9	Calcium-dependent protease, linked to digestive tract function and oxidative stress; located in cytoplasm and nucleus.
LOC101074542	Involved in hormone and progesterone metabolism, binds heme/iron; located in cytoplasm.	LOC115248888	Involved in long-chain fatty acid metabolism; located in endoplasmic reticulum, membrane, and mitochondrion.
LOC101063073	Regulates transcription and circadian rhythm, located in the nucleus.	rhcg2	Supports amino acid metabolism by regulating nitrogen levels, enables ammonium channel activity; located in plasma membrane.
LOC115250540	Defense against bacteria and involved in iron homeostasis; located in the extracellular region.	LOC101078510	L-serine and threonine catabolism. Enables pyridoxal phosphate binding.